

CSCI 3052U: Machine Learning I — Milestone 2: Dataset and Problem Approval Memo

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To: CSCI 3052U instructional team · October 2, 2026 · Repository: [CantoNLU](#)

1. Request

We request approval for the datasets and project description below. The project was obtained from the Lee Language Lab project list (Project 2: Cantonese NLU benchmarking — humans and LLMs), as said at Milestone1, so we also ask that our eligibility for the Lee Lab bonus be confirmed. The lab's team lead for this project, Hyun Min, confirmed by email (September 23, 2026) that we are given permission to use the CantoNLU datasets and the code, which are released under CC by 4.0.

2. Problem, Intended Users and Success Criterion

Problem: How close do naive and classical machine-learning models come to published pretrained-model results on two CantoNLU tasks, language detection and sentiment analysis?

Intended users: Firstly the lee language lab and CantoNLU researchers, who need a simple, reproducible reference layer below their pretrained model, LLM and human results. The second intended users will be practitioners working on low-resource languages with limited compute, for whom a cheap classical model may be good enough.

Success criterion: For each task, the best classical model exceeds the naive benchmarks' validation macro-F1, and its gap to the published CantoNLU scores is reported together with variability across seeds.

3. Data Source, License and Privacy

- Source: the CantoNLU repository (github.com/Aatlantise/canto-nlu) and paper (Min et al., 2025, arXiv:2510.20670).
- License: CC BY 4.0, confirmed by the authors; all derivative work is attributed to them (README, "License and Attribution"). Our own code is MIT-licensed.
- The data is not redistributed in our repository; data/README.md gives the exact download and copy commands.
- Privacy: the sentiment data uses publicly posted restaurant reviews with no reviewer name or ID field; a few review texts contain URLs, restaurant phone numbers or social-media handles, which will not be shown in reports. No data is collected from human participants.

4. Tasks and Data

	Sentiment analysis	Language detection
Unit of analysis	One OpenRice restaurant review	One sentence
Input → output	Review text → smile / ok / cry	Sentence → Cantonese (0) / Mandarin (1) / code-mixed (2)
Official splits (train / validation / test)	9,999 / 999 / 999	42,753 / 2,425 / 2,341
Class balance	Exactly balanced (one third each)	About 16% / 16% / 67%
Missing or empty values	None	None
Published F1 (four CantoNLU models)	70.1–71.9	73.3–78.7
Sample input → output	環境好,食物好,正!!! ... ("Good atmosphere, good food, great!") → smile	建生邨屬於租者置其屋計劃嘅屋邨... → 0 (Cantonese); the Mandarin version differs only in a few characters (嘅/的, 咗/了, 係/是)

More examples with translations are in `DATA_CARD.md` ("Sample Inputs and Outputs").

5. Initial EDA Findings

- **Sentiment** (notebooks/sentiment_eda.ipynb): no missing values, no duplicates within splits, no overlap between splits, exactly balanced classes, and a few very long reviews (up to 4,720 characters).
- **Language detection** (notebooks/language_detection_eda.ipynb): splits are grouped by source sentence, so no source sentence is shared between splits; the classes differ by only a few characters (Cantonese vs. Mandarin similarity 0.88 in an inspected group); 161 training sentences (18 validation, 12 test) appear with two labels, all involving the code-mixed class; 7 train/validation and 8 train/test sentences are repeated across splits; some Mandarin text shows errors from automatic conversion to traditional characters.
- **Task decision:** language detection keeps all three classes, as defined in the CantoNLU paper, and the released data is used unchanged so that results remain comparable with the published ones. The issues above are reported as limitations; a two-class version may be run as an ablation.

6. Train / Validation / Test Strategy

Both tasks use the official CantoNLU train/validation/test splits unchanged (sentiment: 9,999 / 999 / 999; language detection: 42,753 / 2,425 / 2,341), so our results stay comparable with the published ones. The language-detection splits are grouped by source sentence, and our EDA verified that no `source_id` appears in more than one split, so the Cantonese, Mandarin and code-mixed versions of a sentence never leak across splits. All model selection and hyperparameter tuning use the validation split only, and the test split is used once, for the final reported results. TF-IDF vectorizers and any other fitted preprocessing are placed inside a learn pipeline, so they are fitted on training data only. We use the fixed seed 3052 throughout and run at least three seeds where results are stochastic, reporting the mean and standard deviation. As a sensitivity analysis, we plot a learning curve by training on 10%, 25%, 50% and 100% of the training data.

7. Primary Metric

Our primary metric is macro-F1, the unweighted average of the per-class F1 scores, chosen for consistency across the two tasks and because it shows how well each class is handled. Accuracy alone would be misleading for language detection, which is imbalanced (about 67% of examples are code-mixed): a majority-class baseline would reach about 67% accuracy but a macro-F1 of only about 0.27. As secondary measures we report accuracy, per-class precision and recall, and confusion matrices. The CantoNLU paper does not state which F1 average it reports, so when comparing with published scores we report macro-, micro- and weighted-F1 side by side.

8. Feasibility and Compute Plan

The datasets contain 11,997 sentiment reviews and 47,519 language-detection sentences across their official train, validation, and test splits. Measurements were collected on a test machine with a 13th Gen Intel Core i7-13700KF processor, 31.8 GB RAM, and an NVIDIA GeForce RTX 4080 SUPER GPU; PyTorch did not detect CUDA, so all benchmarks ran on the CPU. Character-level TF-IDF with Multinomial Naive Bayes took 6.0 seconds and 622 MB RAM for sentiment analysis and 1.8 seconds and 348 MB for language detection. Logistic regression took 34.3 seconds and 667 MB for sentiment analysis and 9.5 seconds and 352 MB for language detection. LaBSE encoded 500 sentiment examples in 35.7 seconds and 500 language-detection examples in 5.2 seconds, projecting to approximately 14.3 minutes and 8.2 minutes for the complete datasets, respectively. These CPU runtimes show that a GPU is not required for the planned experiments. The 768-dimensional float32 embeddings will be computed once and cached, requiring approximately 180 MB for about 59,500 examples. The environment used Python 3.10.7, NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, Matplotlib 3.10.9, Jupyter 1.1.1, and sentence-transformers 6.1.0.

9. Main Risks

- Label conflicts and synthetic, automatically converted text in language detection may cap scores or let models learn artefacts → reported as limitations and checked in the error analysis.
- The released language-detection data does not match the paper's counts, and the paper does not specify its F1 average → the comparison with published scores is presented as indicative, with three F1 variants.
- Test-set leakage and team time pressure → locked test split, internal deadlines 48 hours early, optional models dropped first.

The full risk register with mitigations is in DATA_CARD.md, Section 5.

10. Changes since Milestone 1 and Evidence

The updated proposal lists the changes since Milestone 1: official language-detection splits exist (no split needs to be constructed), the three-class task definition, the confirmed license, the strengthened metric justification, and a measurable success criterion.

Evidence in the repository: DATA_CARD.md, data/README.md, notebooks/sentiment_eda.ipynb, notebooks/language_detection_eda.ipynb, README.md, LICENSE, CONTRIBUTIONS.md, and AI_USE.md (all AI assistance used for this milestone is logged there).